

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Customer Segmentation Report

Executive Summary

- **The Problem:** A global electronics retailer experience a big drop in Revenue(49.1%) after growth, peaking at \$18.3M in 2019.
- **The Project goal:** identify distinct customer groups based on purchasing behaviour to support targeted marketing and retention strategies.
- **Data:** orders from 2016 to Feb 2021 across 8 countries (online and in-store). Analysis reference date: 31 March 2021.

15,266

Registered
Customers

2,517

Products

26,326

Distinct Orders

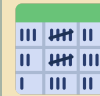
67

Stores

Key Metrics:



Recency: Months since last order



Frequency: Orders per month (relative to customer tenure)



Monetary: Total Spending (Revenue)

Data Processing:



MySQL: data cleaning and RFM segmentation



Power BI: building visualization and interactive reporting

Segment

▼
Whale
VIP
Potential Loyal
New Customer
Loyal Customer
Lapsing
Hibernated
Big Spender
At Risk

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

country

All

Product Categ...

All

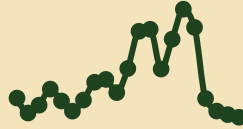
order_type

All

Overview

55.76M

Total Revenue



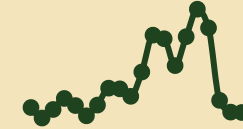
11887

Active Customers



26326

Orders



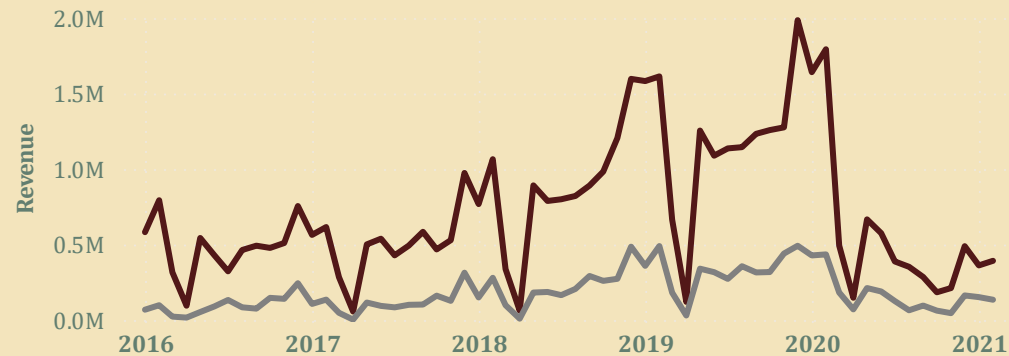
2,117.89

Average Order Value



Revenue by Order Type

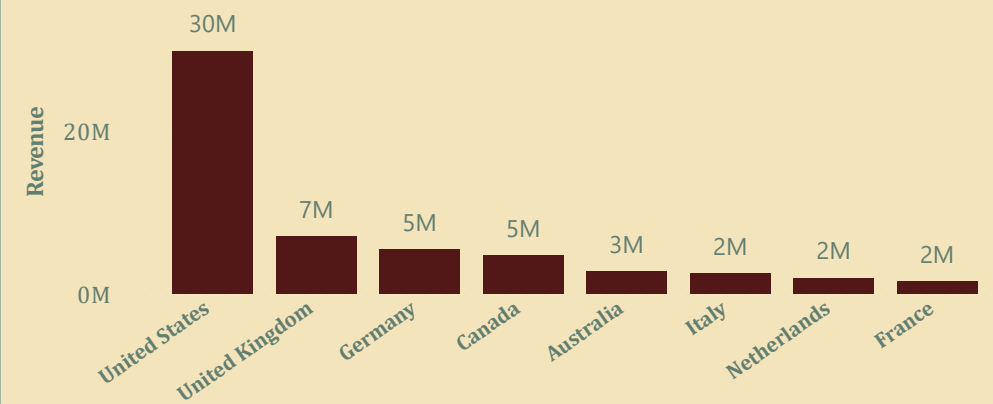
order_type ● In-store ● Online



- **Revenue** peaked at \$18.3M in 2019, dropped 49.1% to \$9.3M in 2020.
- **New customer** acquisition dropped in 2020, driving **Revenue** to drop. **Returning customers**, while also drop, remain relatively more stable.
- **US** contributes ~54% of total **revenue**, largest single market.
- **In-store** dominates revenue but **Online** showed more resilience during 2020.

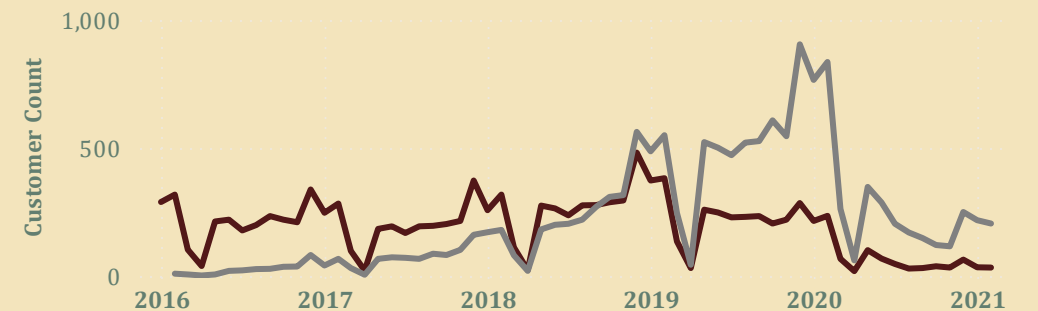
2016 2017 2018 2019 2020 2021

Revenue by Country



New/Return Customer count

Type ● New ● Return



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

country

All

Year

All

Order Type

All

Demographic

28.33M

Male Revenue

6029

Male Count

27.42M

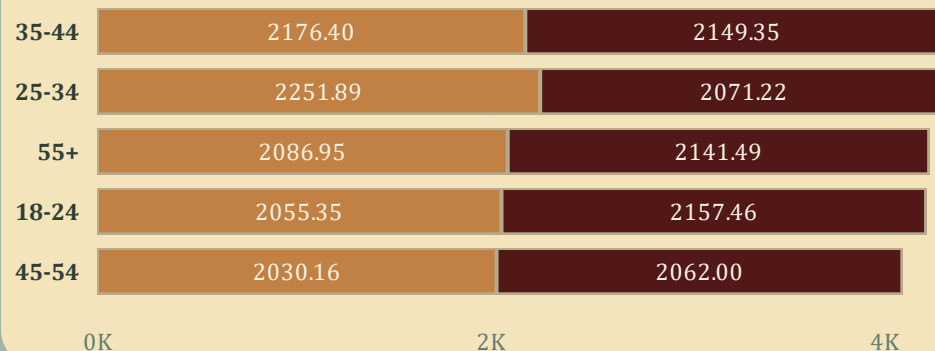
Female Revenue

5858

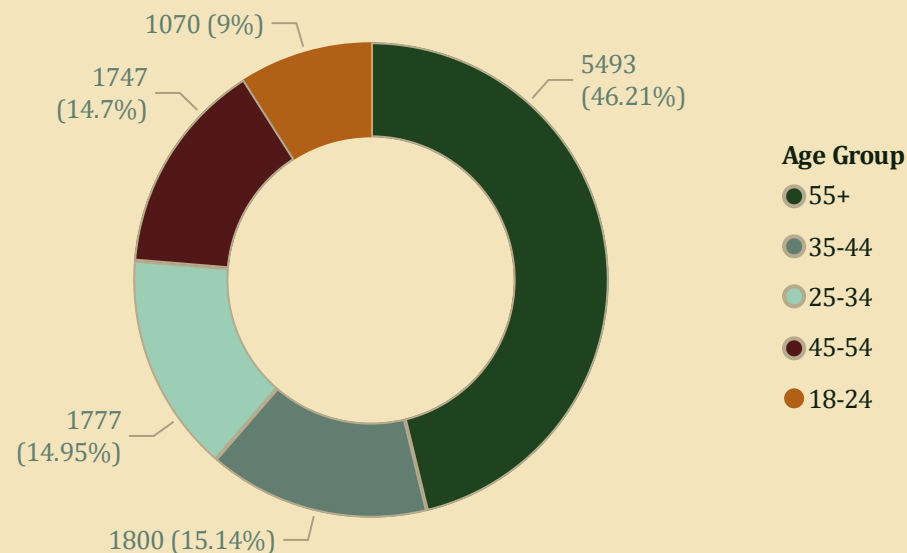
Female Count

AOV by Age Group

Age Group ● Female ● Male

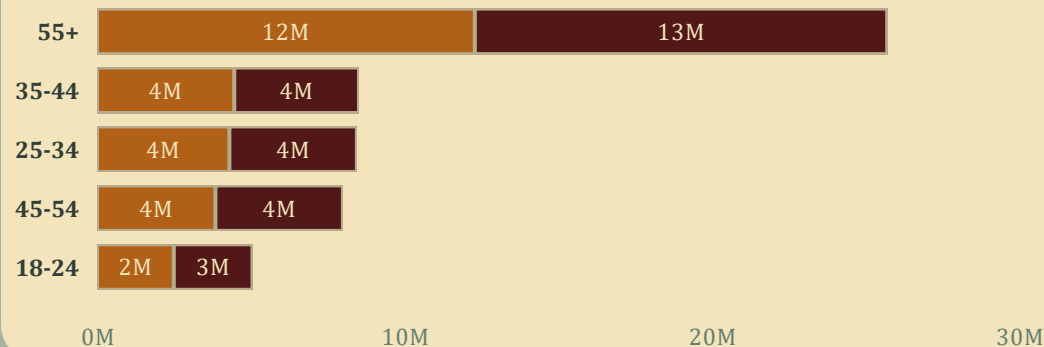


Customer count by Age Group



Revenue by Age Group

gender ● Female ● Male



- **55+** group generate ~**46%** total **Revenue** while also being the biggest group, but their **AOV** does not dominate => leading through volume.
- **Gender** split nearly equal across **Revenue** and **Count**.
- **18-24** group have **highest long-term growth**, but currently the **smallest** group.
- **AOV** showing similar spending behaviors across **Age Group**, **acquisition** is likely the cause of difference in **Revenue**.

[Introduction](#)[Overview](#)[Demographics](#)[Market](#)[Product](#)[Segment Analysis](#)[Retention](#)[Strategy](#)

Country

All

Gender

All

Year

All

Order Type

All

Market Analysis

Market Comparison

All

Italy

11887

Customer count

1130

55.76M

Total Revenue

8.00M

26326

Order Count

1101

2,117.89

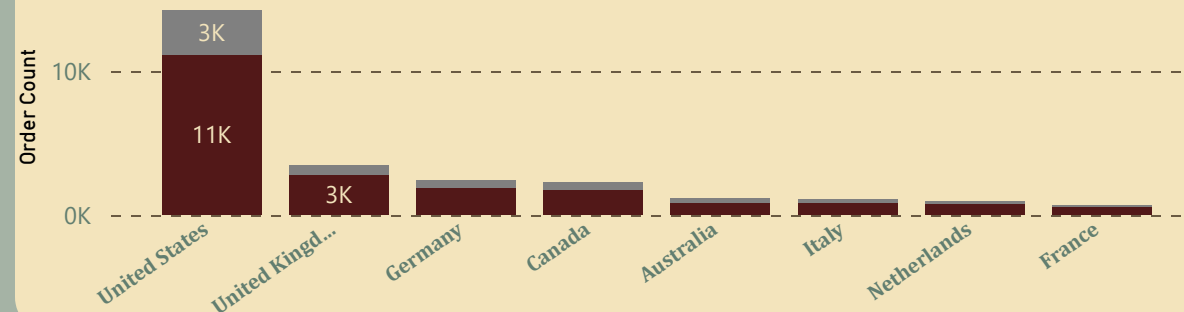
AOV

2,248.54

- **US:** highest **Order Volume** & **Return Rate (40.67%)** but **AOV** sits below average => Most **Loyal Market** lead through **volume**.
- **Aus** and **Italy:** biggest drop in **2019-2020** (~50-53%) in all aspect.
- **France:** smallest drop in both **new (-48%)** and **returning (-10%)** customers during the Pandemic.
- **Online order:** minimal across markets => **underutilized**.
- **New Customers** dominate => **Retention** is key.
- **UK** and **Germany:** perform best in **Europe** best **return**

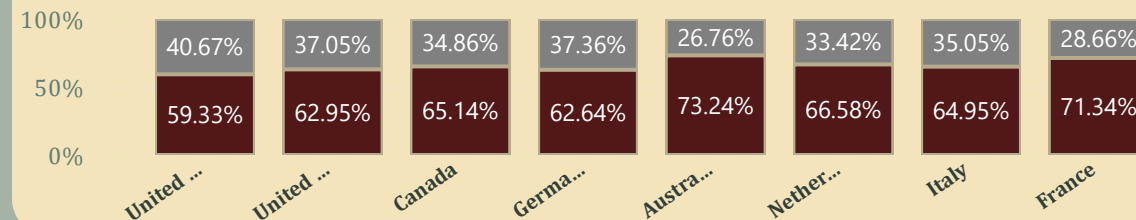
Orders Count by Country

order_type ● In-store ● Online



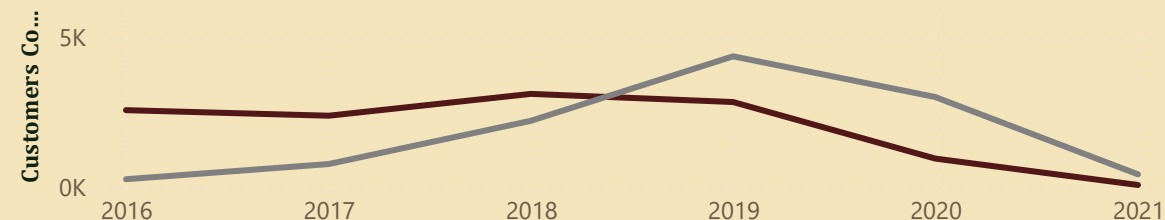
Customer Distribution by Country

Type ● New ● Return



Customer by Type

type ● New ● Return



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Year

All

New/Return

All

Order Type

All

Category

All

Product Analysis

Category Comparison

Cell phones

TV and Video

6253

Customer count

2927

37.35M

Total Revenue

21.52M

8442

Order Count

3325

732.50

AOV

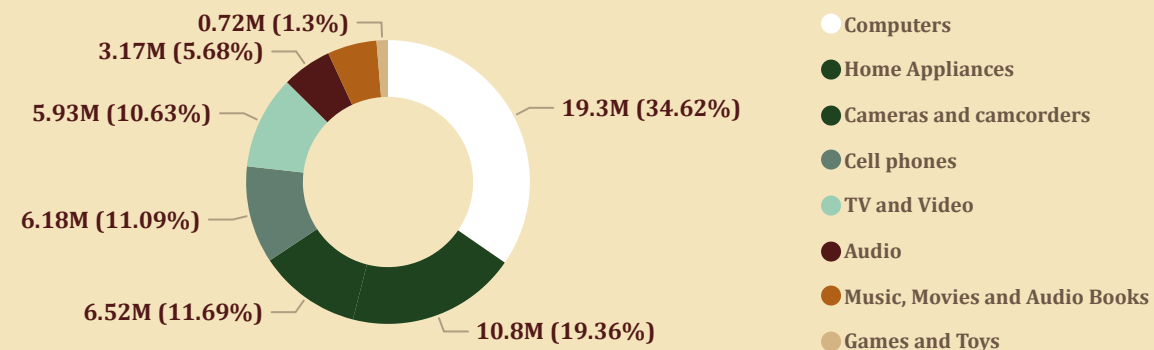
1,783.15

- **Computers** dominates **Revenue (\$19.3M)**, but has average **profit margin (~54%)**.
- **Music, Movies & Audio Books** and **Cameras** have the **highest margins (~57%)** but significantly lower revenue.
- **Home Appliances** generates **2nd highest Revenue (\$11M)** with the **2nd lowest Margin => low efficiency**.
- **Games and Toys** is the weakest => **lowest priority**.
- **All categories** decline in **2020** => **systemic issue**, not **category-specific**.

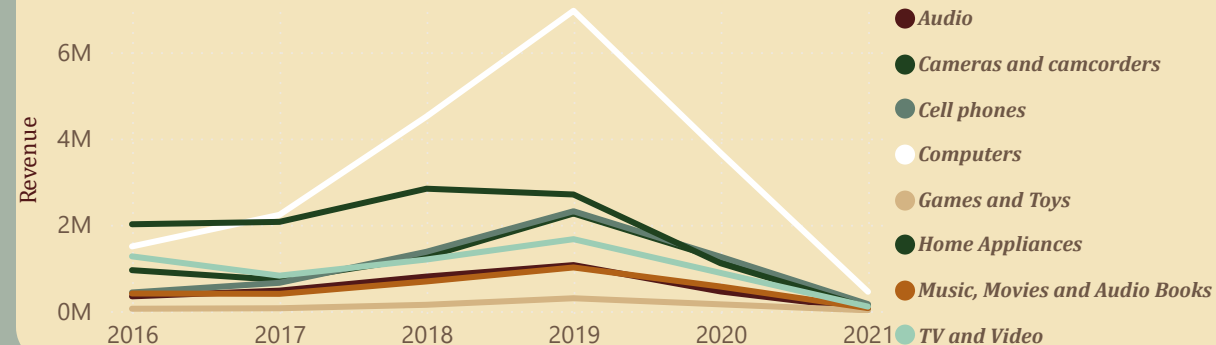
Revenue vs Average Profit Margin



Revenue by Product Category



Revenue by Category over time



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Country

All

Age Group

All

Gender

All

Segment

All

Segment Analysis

Segment Comparison

All

All

11887

Customer
count

11887

55.76M

Total
Revenue

55.76M

4,690.46

Average
Spending

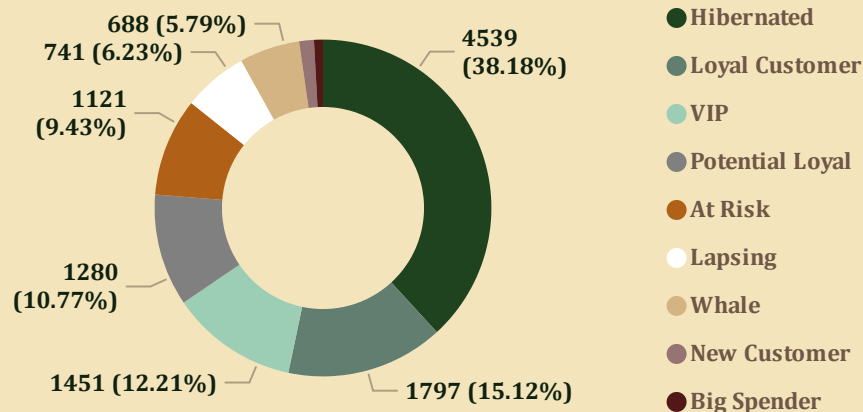
4,690.46

100.00

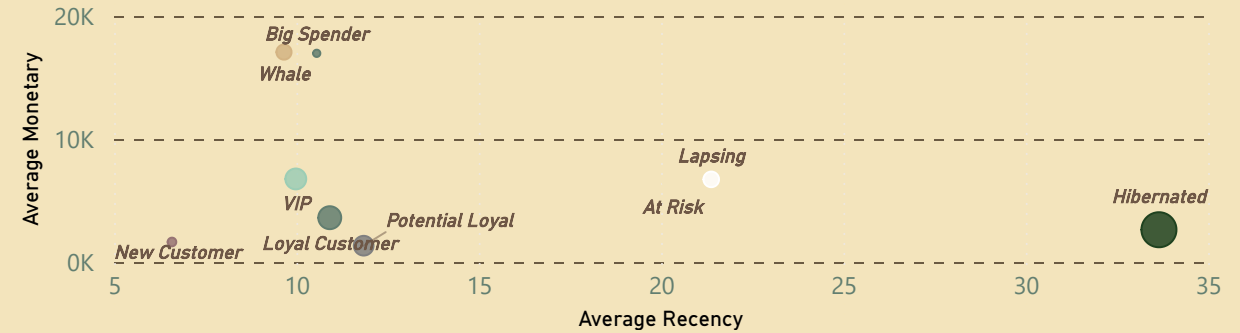
% of total
Revenue

100.00

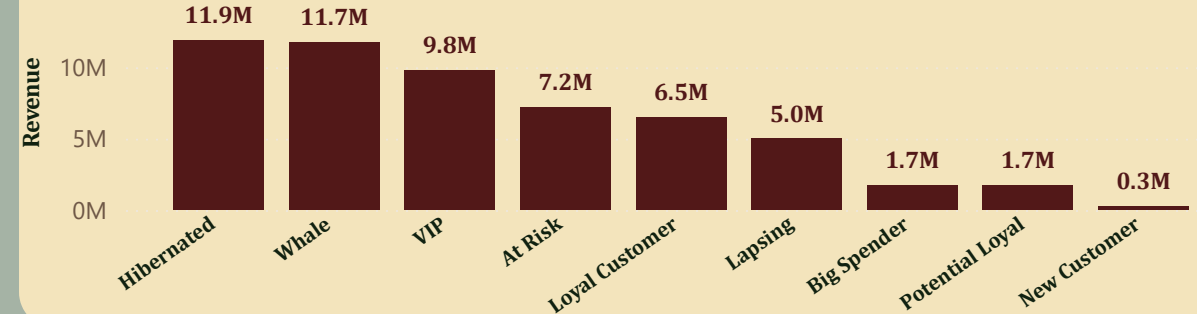
Customers by Segment



Segments Profile Matrix



Revenue by Segment



- **Whale & VIP: 49% Revenue** from only **18% of customers**.
- **Hibernated** generated the **2nd highest Revenue (\$11.7M)** => high value to recover.
- **At Risk** generates more **Revenue** than **Loyal Customers** with less customers => Higher **individual value**, **urgent to recover**.
- **New customer** is the smallest with the least **Revenue**, signaling slow **customer acquisition rate**.

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

year

All

month

All

Segment

All

Month After

All

Retention Rate

3.21

Avg 1 month Retention Rate

7,167

Post-1st-Month Returner

17.39

Peak Retention Rate

199.85

Avg Monthly New Customer

Return Customer by month

Year	1	2	3	4	5	6	7	8	9
⊕ 2016	55	42	39	47	48	47	46	40	52
⊕ 2017	33	53	45	44	74	60	61	69	73
⊕ 2018	120	126	130	112	124	133	169	135	163
⊕ 2019	147	116	126	131	123	127	130	113	113
⊕ 2020	27	5	6	12	11	8	5	10	8
Total	382	342	346	346	380	375	411	367	409

Cohort Retention Rate (%) by month

Year	1	2	3	4	5	6	7	8
⊕ 2016	2.15	1.64	1.52	1.84	1.87	1.84	1.80	1.5
⊕ 2017	1.39	2.23	1.89	1.85	3.11	2.53	2.57	2.5
⊕ 2018	3.87	4.06	4.19	3.61	3.99	4.28	5.44	4.2
⊕ 2019	5.19	4.10	4.45	4.63	4.34	4.48	4.59	3.5
⊕ 2020	2.85	0.53	0.63	1.27	1.16	0.84	0.53	1.0
Average	3.21	2.88	2.91	2.91	3.20	3.15	3.46	3.0

- **Average retention** of 3.21% after 1 month => ~97% of customers do not return the following month after their first purchase.
- **Retention Rate** varies significantly (~0-17%), suggesting **sporadic repurchase behavior** rather than consistent loyalty.
- **2019** has the best **1-month retention** (5.19%), while it collapse in **2020** (2.85%) and drop significantly (~0-1%) => **Customer** almost never come back after their 1st purchase of the year.
- **At Risk:** Widest **Average retention rate** range (~3.39-9.55), peak **retention** rivals the highest performers while their floor still exceeds most **segments**. => **6 months** after is the **critical** time for intervene before they become **Hibernated**

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Recommendation

PROTECTING

Whale, VIP and Loyal Customers:

- Exclusive offer, early access/release.
- Set up **Recency** warning, personal reach-out after about every **2-3 months**.
- Target **55+** group in the segments, especially **Male Whale**.

US Market: Prioritize for protecting. target all segments here.

Computer and Home appliances: Most sales with decent profit, promote sales/offers for new release and upgrades.

WIN BACK

At Risk:

- highly reactive within **6 months** from last order (~**7-9% Retention Rate**) => best time to win back. have **gender-specific** offer, target **Female** (more **Revenue**)
- Lapsing** spend more, less often, push for **Category** relevance.
- Australia and Italy:** steepest **customer losses** in **2020** — **highest priority** markets for **win back** campaigns.

REACTIVATING

Hibernated:

- Target the recently Hibernated based on average **Recency**(last active within ~**25 months**).
- Lead with **New drops** since they have decent **average spending**, just long **inactivity**.

Online: better chance to reach **dormant Customers**.

Music/Movies and Cameras: Highest **Profit Margin** (~**56-57%**) — good fit for **reactivation campaign** with.

DEVELOPING

New and Potential Loyal.

- ~**97%** of **New Customers** don't come back after **1 months** => leverage the **1st 90 days**.
 - **Potential Loyal** are active quite **recent** but not committed => **offers** or **loyalty programs** invite
- Online:** utilize more to acquire and keep more **New Customers**, especially for younger **Demographics**.
- Australia:** Highest proportion of **New Customers** across **all years** => Best market for **conversion campaign**.